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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

HACKATHON 2026

Idea Submission Template — Code. Create. Innovate.

Team Name:

Team Raksha

Problem Statement No:

2

Problem Statement Title:

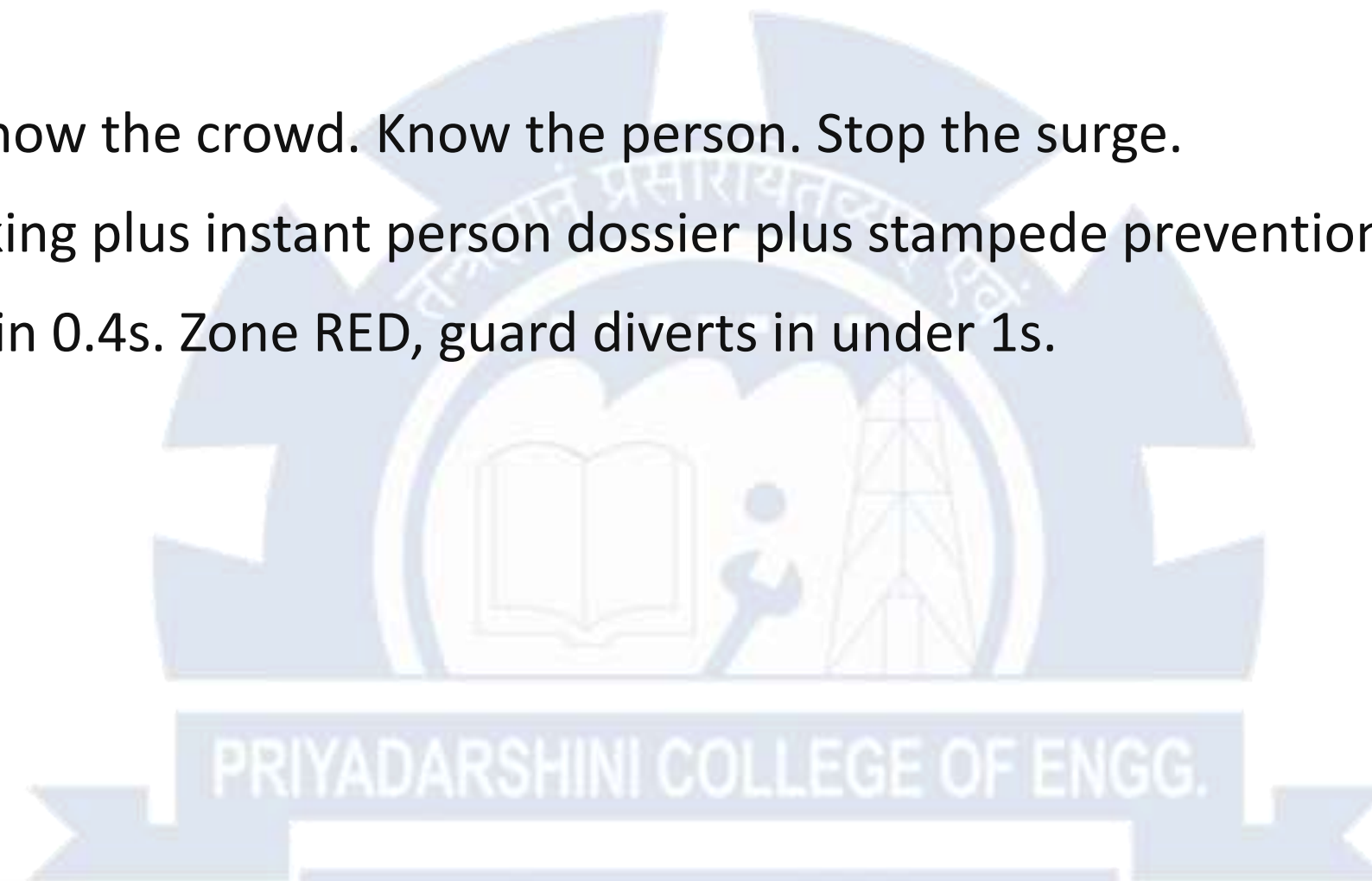
Improving Safety at Large Public Events

Team Leader Name:

[Your Name]

IDEA TITLE

- **RAKSHA Command Center:** Know the crowd. Know the person. Stop the surge.
- **What:** Live multi-camera tracking plus instant person dossier plus stampede prevention
- **Wow:** Face appears, file pops in 0.4s. Zone RED, guard diverts in under 1s.



Live Tracking

YOLOv8 plus ByteTrack IDs survive occlusion and cross-cam ReID

Instant Dossier

ArcFace 512-D plus age, gender, clothing and trajectory map

Proactive Action

Density times stillness times counterflow equals risk meter

TECHNICAL APPROACH

- **Ingest:** RTSP, webcam, video file plus zone polygons. 720p and frame-skip 2 for CPU speed.
- **Detect plus Track:** YOLOv8n plus ByteTrack persist. OSNet ReID plus Hungarian for global ID.
- **Face plus Attributes:** Best-snapshot scoring for blur and pose. ArcFace plus clothing and age.
- **Search plus Analyze:** Faiss one-person-one-file. Density, flow and bottleneck alerts under 1s.

Pipeline

Ingest to Track to Face to Match to Analyze to Act

Quality Gate

Only sharp frontal faces enroll. No junk matches.

Demo

CAM-03: 42 persons, A-1047 at 0.91, Zone C RED, Exit 2

FEASIBILITY AND VIABILITY

- **Tracking measured:** YOLOv8s plus ByteTrack 15 to 30 FPS on RTX 4050, 8 with full overlay. Assoc under 1ms.
- **Face measured:** buffalo_s 61ms steady on GPU, was 438ms on CPU. Faiss gallery instant under 50k IDs.
- **Stampede warning:** AURORA 94.3 percent, 13.8s early lead. Our rule meter fired RED 1.2 plus on dense test.
- **24h MVP:** Webcam plus clips plus enrolled faces. No extra hardware needed for demo.

Evidence

2025-26 CPU studies plus InsightFace benchmarks

Tuning

640 input, 720p, skip 2 frames for real-time laptop

Fallback

IoU tracker 64 FPS and clothing search if face hidden

IMPACT AND BENEFITS

- **Lives saved:** Bottleneck caught 60s before human. Early diversion with PA and signage.
- **Find fast:** Lost kid or suspect by photo or clothing filter in under 2s with clip export.
- **Where it fits:** Mela, stadium, station, campus and mall. Same wall, zone-wise push.
- **Pilot metric:** Over 60s lead time, under 2s search, under 1s alert, audit ready.

Organizers

One screen replaces 50-screen stare

Police

Dossier plus trail plus audit in one click

Public

Safer exits, faster help, blurred storage

RESEARCH AND REFERENCES

- **YOLO plus ByteTrack:** 2025-26 studies: YOLOv8n best balance. ByteTrack 14 percent faster than BoT-SORT.
- **ArcFace plus Faiss:** Buffalo pack 99.5 percent LFW. Cosine 0.40 threshold. Temporal smoothing stabilizes.
- **Stampede models:** AURORA hybrid plus FTLE bottleneck detection, AVSS 2019. Delhi pilot cut response 37pc.
- **India context:** Hathras, Kumbh and station pattern: density plus blocked exit. STAR-Crowd Kumbh data.

